

ENV 6106

Air Pollution Modeling

Spring 2023

Topics

- CAL3QHC v2.0
- Expectation
 - Understand the basic concepts and input file structure
 - Be able to make an input file and run CAL3QHC model
- Reference:
 - CAL3QHC user guide:
 - <https://www.weblakes.com/products/calroads/resources/docs/CAL3QHC.pdf>
 - CO Florida 2012 Report:
 - <http://www.fdot.gov/environment/software/COFL2012%20Final.pdf>

CAL3QHC Basics

- A steady-state Gaussian dispersion model
 - Same model type as AERMOD, much simplified
- CALINE3 + an intersection queuing algorithm
 - CALINE3: California Line source model 3rd gen
 - Developed by California DOT for roadways
- Designed for free-flow links and Signalized intersections, and inert pollutants
- Model roadways as a series of “line” sources

CAL3QHC Basics

- Use for CO analysis
- Not to be used for PM hot-spot analysis
- Up to 120 roadway links, 60 receptor locations, and 360 wind angles
- Does not take hourly meteorological data
 - CAL3QHCR could

CAL3QHC control file

- CAL3QHC reads a control file, which provides all data/information needed for modeling
- CAL3QHC uses three formats to read control file
 - Character: text string, need to be single quoted
 - E.g: 'this is a test link 1 2 3'
 - Integer: number with no decimal points
 - Real: number with decimal points

Control file: Line 1

Name	Type	Description
JOB	Character	Job title, < 40 character
ATIM	Real	Averaging time (min). 60 in most case
ZO	Real	Surface roughness length (cm)
VS	Real	Settling velocity (cm/s). How fast particle is settling. Usually 0
VD	Real	Deposition velocity (cm/s). "Equivalent" velocity for dry and wet deposition. Usually 0
NR	Integer	# receptor, max = 60
SCAL	Real	Scaling factor to convert input number to meter. 1 if inputs are in meter. 0.3048 if inputs are in feet
IOPT	Integer	1 output feet; 0 output meter
IDEBUG	Integer	1 for more detailed diagnostic information. 0 disable

Control file: Line 1

Example 1

'EXAMPLE - TWO WAY INTERSECTION (EX-1)' 60. 175. 0. 0. 8 0.3048 1 1

- The name of job is “EXAMPLE - TWO WAY INTERSECTION (EX-1)”
- Averaging time is 60 minutes = 1 hour
- Surface roughness length is 175 cm (should be urban region)
- No settling, no dry and wet deposition
- 8 receptors
- Input numbers are in feet, times 0.3048 to convert to meter
- Output numbers in feet, and display diagnostic information
- *Note that input number for “real” type always have a dot*

Control file: Next NR lines

Name	Type	Description
RCP	Character	Name of this receptor, not a group of receptor
XR	Real	X-coordinate of this receptor
YR	Real	Y-coordinate of this receptor
ZR	Real	Z-coordinate. Usually set to breathing height around 1.8 m

Example 1

'REC 1 (SE CORNER) '	45.	-35.	6.0
'REC 2 (SW CORNER) '	-45.	-35.	6.0
'REC 3 (NW CORNER) '	-45.	35.	6.0
'REC 4 (NE CORNER) '	45.	35.	6.0
'REC 5 (E MID-MAIN) '	45.	-150.	6.0
'REC 6 (W MID-MAIN) '	-45.	-150.	6.0
'REC 7 (N MID-LOCAL)'	-150.	35.	6.0
'REC 8 (S MID-LOCAL)'	-150.	-35.	6.0

Control file: Next 1 line

Name	Type	Description
Run	Character	Title of run detailed. < 40 character
NL	Integer	# links. Max = 120
NM	Integer	# meteorological conditions. No # limits
PRINT2	Integer	1 for detailed output, 0 for summary
MODE	Character	'C' for CO; 'P' for PM. Single quote

Example 1

'MAIN ST. AND LOCAL ST. INTERSECTION' 9 1 0 'C'
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- The name of this model run is “MAIN ST. AND LOCAL ST. INTERSECTION”
- There are 9 links modeled, only 1 meteorological condition
- Print summary
- Run model for CO

Control file: Link data line 1

Name	Type	Description
IQ	Integer	1 for free-flow link and 2 for queue links

- Free-flow links (**1 line of extra data needed**)
 - Where driver could travel without delay
 - Usually one end at center of intersection
- Queue links (**2 lines of extra data needed**)
 - Where vehicle will idle with speed = 0
- Links should cover all roadway section.
 - E.g.: end of one link = start of another
- Recommend include all roads in 1000 ft range

Free-flow link (link data line 2)

Name	Type	Description
LNK	Character	Link description. Max = 20 characters
TYP	Character	Link type. Can be: AG, FL, BR, DP
XL1	Real	X-coordinate of end point 1
YL1	Real	Y-coordinate of end point 1
XL2	Real	X-coordinate of end point 2
YL2	Real	Y-coordinate of end point 2
VPHL	Real	Traffic volume (vehicle per hour)
EFL	Real	Emission factor (grams per vehicle mile)
HL	Real	Height of road, should be within + - 10 m
WL	Real	Mixing zone width, should be link width plus 3 m each side

Free-flow link

- Link type
 - **AG**: at-grade, a normal intersection where road intersect in the same height. Most cases use this
 - Others: elevated road intersection (overpass)
 - **FL**: fill, filled under elevated road, leaving one tunnel
 - **BR**: bridge, not filled under elevated road, wind passes freely
 - **DP**: depressed, road dipped below under elevated road
- Link length should > link width

Example 1

1
'Main St.NB Appr. ' 'AG' 10. -1000. 10. 0. 1500. 41.6 0. 40.

- Free-flow link, at-grade (same height as other roads). 0 m height
- Volume: 1500 vehicles/hour. Emission factor: 41.6 g/mile
- Width: 40 ft (actual width is ~20 ft, plus ~10 ft each side)

Queue link (link data line 2)

Name	Type	Description
LNK	Character	Link description. Max = 20 characters
TYP	Character	Link type. Can be: AG, FL, BR, DP
XL1	Real	X-coordinate of end point 1
YL1	Real	Y-coordinate of end point 1
XL2	Real	X-coordinate of end point 2
YL2	Real	Y-coordinate of end point 2
HL	Real	Height of road, should be within + - 10 m
WL	Real	Mixing zone width, should be link width, no plus 3 m
NLANES	Integer	# travel lanes in queue link

Queue link (link data line 3)

Name	Type	Description
CAVG	Integer	Average total signal cycle length (s)
RAVG	Integer	Average red total signal cycle length (s)
YFAC	Real	Clearance lost time. Portion of the yellow phase that is not used by motorist (s).
IV	Integer	Approach volume on queue link (vehicle per hour)
IDLFAC	Real	Idle emission factor (grams per vehicle hour)
SFR	Integer	Saturation flow rate (vehicle / hour / lane). If 0 set to default: 1600
ST	Integer	Signal type. 1: pretimed; 2: actuated; 3: semi-actuated. If 0 set to default: 1
AT	Real	Arrival rate. 1 through 5. If 0 set to default: 3

Queue link

- Link length: only used to determine direction for queue link, actual queue length estimated internally
- Signal type
 - 1: pretimed: fixed signal time length
 - 2: actuated: all signals triggered as needed
 - E.g.: loop-detector or push-button
 - 3: semi-actuated.
 - Some signal triggered as needed, some not
- Arrival rate
 - Traffic signal time is designed to optimize traffic flow
 - Best case: drive through several intersections without stop
 - Worst case: run into every red light at every intersection

Queue link

- Arrival rate
 - 1 = Worst: cars lined up at beginning of red
 - 2 = Below ave: cars lined up in middle of red
 - 3 = Average: random arrivals
 - 4 = Above ave: dense cars arrives during green
 - 5 = Best: dense cars at beginning of green

Example 1

2
'Main St.NB Queue ' 'AG' 10. -10. 10. -1000. 0. 20.0 2
90 40 3.0 1500 735.00 0 0 0

- Queue link, at-grade, 0 m height, 20 ft actual width with 2 lanes
- Total signal length 90 sec, 40 sec read light, 3 sec lost to yellow
- Approach volume 1500 vehicle/h/lane with EF 735 g/vehicle-h
- SFR, ST & AT are zero → set to defaults

Control file: Meteorology data

Name	Type	Description
U	Real	Wind speed (m/s). Use 1 m/s for worse case
BRG	Real	Wind direction. If 0: wind direction varies (VAR = 'Y')
CLAS	Integer	Stability class. A – F: 1 – 6
MIXH	Real	Mixing height (m). Default: 1000 m
AMB	Real	Ambient background concentration (ppm for CO)
VAR	Character	'Y': wind direction varies; 'N': no direction variation
DEGR	Integer	Wind direction increment angle (degrees)
VAI(1)	Integer	Lower bound of variation. Start angle = VAI(1) * DEGR
VAI(2)	Integer	Upper bound of variation. End angle = VAI(2) * DEGR

Control file: Meteorology data

- Stability class
 - Can use D (4) for urban region and E (5) for rural region
 - Can also assume F (6) for absolute worst case

Example 2

1.0	00.	4	1000.	0.	'Y'	5	30	42
1.0	00.	4	1000.	0.	'Y'	3	80	100
2.0	00.	4	1000.	0.	'Y'	10	33	43

- Three sets of meteorology data
- 1st set: 1 m/s, spatial varying wind dir, stability class D, 1000 m mixing height, no background concentration, wind dir from $5 \times 30 = 150$ to $5 \times 42 = 210$ degree
- 2nd set: similar with 1st set, but wind dir from 240 to 300 degree
- 3rd set: similar with 1st set, but with wind at 2 m/s and wind dir from 330 to 430 (70) degree

Where to place receptor

- Along areas impacted by the project
- At locations where high concentration expected
 - Next to road, but outside mixing zone
 - All directions from the project
- No receptor at locations restricted to public
- Also place receptor up to a few hundreds meters away from the project (e.g.: 500 m)
- Receptor spacing denser near project (e.g. 25 m), and wider further away (e.g.: 100 m)

Queue links special case

- # vehicles in queue and queue length estimated internally by CAL3QHC
- When over-capacity ($V/C > 1$), queue length may longer than physical length
 - Check output page 1 V/C number
- Can model queue link as free-flow link, with EF set to 100 g/vehicle-mile and volume set to estimated VPH
 - See CAL3QHC v2.0 user guide section 3.3.2 & 4.2 for more information